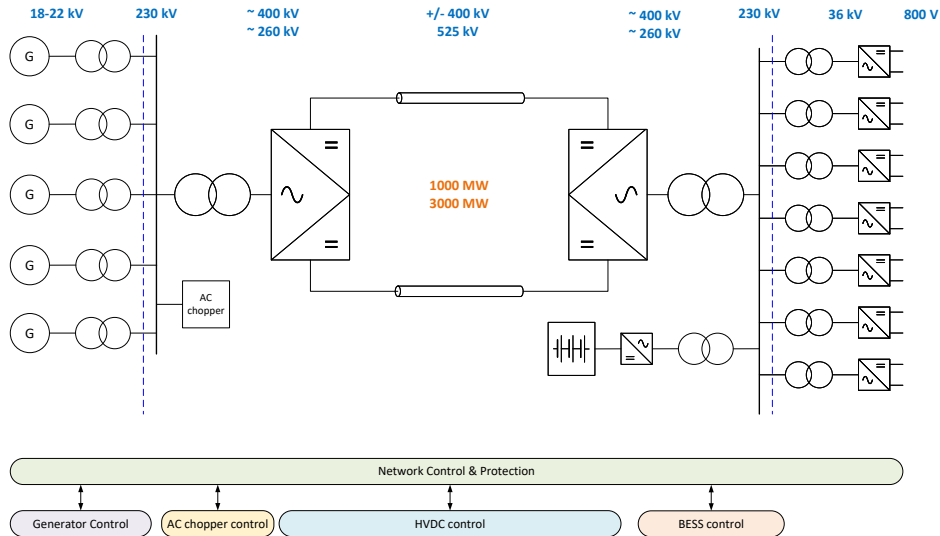


AMERICAN TERAWATT

Utah and Nevada HVDC projects

System Requirements

System configuration



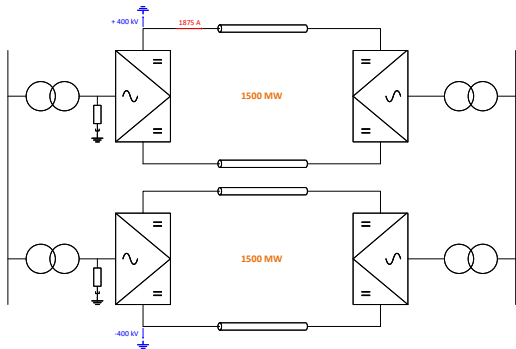
System configuration

- **Power** - Projects with $1000MW$ and $3000MW$ power transmission requirements
- **DC line** - 130 -200 miles, with priority to OHL, but cable sections can be implemented to accelerate the permission process.
- **Generation** - Geothermal power plants is the main generation source. Additional generation sources might be implemented once the transmission system is established.
- **Load** - The main load are data centers.
- **BESS** - Battery storage is assumed for backup and for datacenter load flicker compensation.
- **AC Chopper** - For fault ride through, installation of the AC chopper might be required.

System configuration

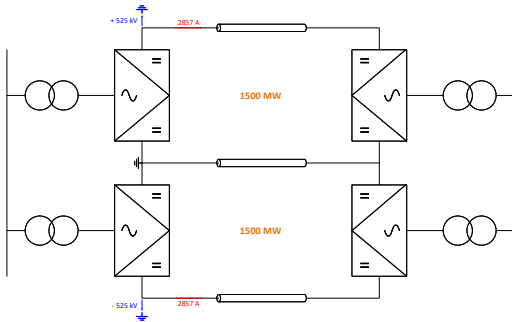
- Islanded operation with no interconnection to the AC grid
- Generation is build out in $300MW$ steps and is defined by the water availability
⇒ Staged buildout of the transmission line is beneficial
- High reliability requirements by datacenters
- Standardized hardware design approach to reduce hardware lead times and configuration variations
- Building up C&P capabilities to reduce risk and execution times in project execution
- **Reduction of execution time is very critical**
- 4 initial projects, 2 x $1000MW$ and 2 x $3000GW$

HVDC Configuration - Dual SMP



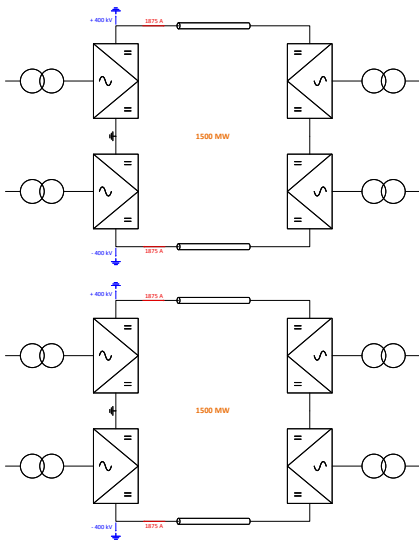
- **Topology** - Dual Symmetrical Monopole with $\pm 400kV$
- **DC Line** - DC cable and OHL can be utilized with the given current rating
- **Stages** - Two $1500MW$ stages
- **Flexibility** - Design can be used for configurations with $1000MW$ rating

HVDC Configuration - BIP



- **Topology** - Bipole with $\pm 525\text{ kV}$
- **DC Line** - OHL can be utilized with the given current rating but parallel cables are required
- **Stages** - Two 1500 MW stages
- **Flexibility** - Limited flexibility as asymmetrical configuration is not economical for most requirements

HVDC Configuration - Rigid BIP



- **Topology** - Dual Rigid Bipole with $\pm 400kV$ and **3ph** transformer units
- **DC Line** - DC cable and OHL can be utilized with the given current rating
- **Stages** - 750MW stages for converter and 1500MW stages for DC line
- **Flexibility** - Limited flexibility as asymmetrical configuration is not economical for most requirements

Working Packages Ongoing

- **DC line**

- Local company is working on DC route definition and DC line design

- **Network Control & Protection**

- Operation concept definition
 - Offline studies for concept verification
 - Development of models for performance design and verification
 - Definition of performance verification tests and evaluation criteria
 - Control hardware definition

Working Packages

HVDC configuration by build size and phasing

- Recommended configuration for 300MW, 500MW, 1GW, 2GW, 3GW
 - Monopole vs bipole by capacity
 - Voltage class by size and distance: $\pm 320kV$, $\pm 400kV$, $\pm 525kV$
- Modular expansion path from first stage to GW scale
 - Oversizing on day 1 vs future expansion
 - Oversizing for redundancy
 - Expansion from 300 – 500MW initial build to 1 – 2GW
 - Design approach for staged customer commitments
 - Cost and schedule penalties from deferred expansion
 - Avoiding stranded capex